

Wireshark

ADVANCED COMPUTER NETWORK LAB

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Wireshark

Network Packet / protocol Analyser

Open Source Network Tool

Live capture and offline analysis

Some examples people use Wireshark for:

- Network administrators: troubleshoot network problems
- Network security engineers: examine security problems
- Developers: debug protocol implementations
- People: learn network protocol internals

Interfaces

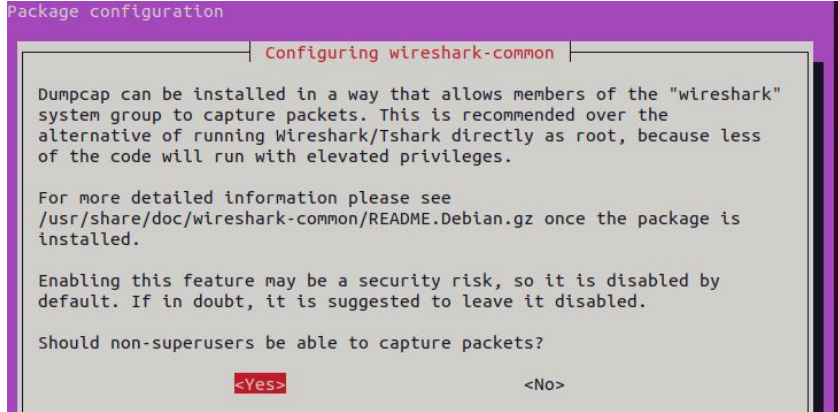
\$ ip addr

Interface	Role	IP
lo (Loopback)	Internal	127.0.0.1
eno1 (Wired - Ethernet)	Main network	10.10.106.78
wlp0s20f3 (Wireless - Wi-Fi)	Hotspot/local	10.42.0.1

Installation

\$ sudo apt update

\$ sudo apt install wireshark



Add user to Wireshark

HINT - whoami

\$sudo usermod -aG wireshark systemslab

Executable permission to dumpcap file

```
$ ls -l /usr/bin/dumpcap
```

```
-rwxr-x--- 1 root wireshark 129696 Mar  4 2022 /usr/bin/dumpcap
```

If NOT

```
sudo chgrp wireshark /usr/bin/dumpcap
```

```
sudo chmod 750 /usr/bin/dumpcap
```

packet capture engine - dumpcap

Check capabilities:

```
$ getcap /usr/bin/dumpcap
```

```
/usr/bin/dumpcap cap_net_admin,cap_net_raw=eip
```

```
$ groups
```

```
systemslab adm cdrom sudo dip plugdev lpadmin lxd sambashare
```

```
$ newgrp wireshark
```

```
$ groups
```

```
wireshark adm cdrom sudo dip plugdev lpadmin lxd sambashare systemslab
```

Color Rules

Name	Filter
☒ Bad TCP	tcp.analysis.flags && !tcp.analysis.window_update && !tcp.analysis.
☒ HSRP State Change	hsrp.state != 8 && hsrp.state != 16
☒ Spanning Tree Topology Change	stp.type == 0x80
☒ OSPF State Change	ospf.msg != 1
☒ ICMP errors	icmp.type eq 3 icmp.type eq 4 icmp.type eq 5 icmp.type eq 11
☒ ARP	arp
☒ ICMP	icmp icmpv6
☒ TCP RST	tcp.flags.reset eq 1
☒ SCTP ABORT	sctp.chunk_type eq ABORT
☒ TTL low or unexpected	(! ip.dst == 224.0.0.0/4 && ip.ttl < 5 && !pim && !ospf) (ip.dst == 2
☒ Checksum Errors	eth.fcs.status=="Bad" ip.checksum.status=="Bad" tcp.checksum
☒ SMB	smb nbss nbns netbios
☒ HTTP	http tcp.port == 80 http2
☒ DCERPC	dcerpc
☒ Routing	hsrp eigrp ospf bgp cdp vrrp carp gvrp igmp ismp
☒ TCP SYN/FIN	tcp.flags & 0x02 tcp.flags.fin == 1
☒ TCP	tcp
☒ UDP	udp
☒ Broadcast	eth[0] & 1
☒ System Event	systemd_journal sysdig

Capture

...using this filter:

eno1

wlp0s20f3

any

Loopback: lo

bluetooth0

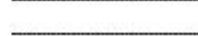
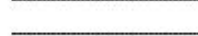
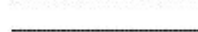
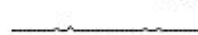
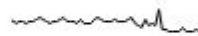
bluetooth-monitor

nflog

nfqueue

dbus-system

dbus-session



1: Packet Identification & Addressing

Questions

1. Identify **any 10 packets** from your capture.
2. For each packet, mention:
 - Source IP address
 - Destination IP address
 - Protocol used
3. Identify the **MAC address** of:
 - Your system
 - The destination device

1. Open Wireshark
2. Select interface:
 - a. wlp0s20f3 (Wi-Fi) OR eno1 (LAN)
3. Click Start
4. Open browser → visit:example.com
5. Let packets capture for ~20–30 seconds
6. Click Stop

- Select any 10 packets (rows)
- For each packet:
- Look at columns:
 - Source
 - Destination
 - Protocol

Wired

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
Apply a display filter ... <Ctrl-/>						
No.	Source	Destination	Time	Protocol	Length	Info
1	172.64.148.235	10.10.106.78	0.0000000000	TLSv1.2	90	Application Data
2	10.10.106.78	172.64.148.235	0.000220456	TLSv1.2	94	Application Data
3	172.64.148.235	10.10.106.78	0.028710682	TCP	66	443 → 56056 [ACK] Seq=25 Ack=29 Win=19 Le
4	10.10.106.78	142.251.221.110	0.161952049	TLSv1.2	186	Application Data
5	10.10.106.78	142.251.221.110	0.161987676	TLSv1.2	1708	Application Data, Application Data
6	142.251.221.110	10.10.106.78	0.184845600	TCP	66	443 → 34828 [ACK] Seq=1 Ack=1521 Win=976
7	142.251.221.110	10.10.106.78	0.185576660	TLSv1.2	105	Application Data
8	10.10.106.78	142.251.221.110	0.225605340	TCP	66	34828 → 443 [ACK] Seq=1763 Ack=40 Win=501
9	142.251.221.110	10.10.106.78	0.332957017	TLSv1.2	136	Application Data
10	10.10.106.78	142.251.221.110	0.332997948	TCP	66	34828 → 443 [ACK] Seq=1763 Ack=110 Win=50

Packet No.	Source IP	Destination IP	Protocol
1	172.64.148.235	10.10.106.78	TLSv1.2
2	10.10.106.78	172.64.148.235	TLSv1.2
3	172.64.148.235	10.10.106.78	TCP

```
▶ Frame 8: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface eno1, id 0
▼ Ethernet II, Src: HewlettP_ba:4a:01 (d8:94:03:ba:4a:01), Dst: 6c:0b:5e:56:55:06 (6c:0b:5e:56:55:06)
  ▶ Destination: 6c:0b:5e:56:55:06 (6c:0b:5e:56:55:06)
  ▶ Source: HewlettP_ba:4a:01 (d8:94:03:ba:4a:01)
    Type: IPv4 (0x0800)
▶ Internet Protocol Version 4, Src: 151.101.105.91, Dst: 10.10.106.78
▶ Transmission Control Protocol, Src Port: 443, Dst Port: 53724, Seq: 79, Ack: 36, Len: 0
```

Source MAC address:

d8:94:03:ba:4a:01 → (*our system*)

Destination MAC address:

6c:0b:5e:56:55:06 → (this is **router / gateway**)

Wireless

No.	Source	Destination	Time	Protocol	Length	Info
1	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.000000000	TLSv1.2	1294	Application Data
2	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.000000288	TLSv1.2	1294	[TCP Previous segment...]
3	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.000000328	TLSv1.2	1294	Application Data
4	2409:40e2:24:...	2404:6800:4009:809::2001	0.000034396	TCP	98	34440 → 443 [ACK] Seq=...
5	2409:40e2:24:...	2404:6800:4009:809::2001	0.000060711	TCP	106	[TCP Dup ACK 4#1] Seq=...
6	2409:40e2:24:...	2404:6800:4009:809::2001	0.000171527	TCP	106	[TCP Dup ACK 4#2] Seq=...
7	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.007378295	TLSv1.2	1294	Application Data
8	2409:40e2:24:...	2404:6800:4009:809::2001	0.007499170	TCP	106	[TCP Dup ACK 4#3] Seq=...
9	2409:40e2:24:...	2404:6800:4009:826::2001	0.036451400	TCP	86	58270 → 443 [ACK] Seq=...
10	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.076571855	TLSv1.2	1294	[TCP Previous segment...]
11	2409:40e2:24:...	2404:6800:4009:809::2001	0.076698200	TCP	114	[TCP Dup ACK 4#4] Seq=...
12	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.080108521	TLSv1.2	1294	Application Data
13	2409:40e2:24:...	2404:6800:4009:809::2001	0.080229876	TCP	114	[TCP Dup ACK 4#5] Seq=...
14	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.089513705	TLSv1.2	1294	Application Data
15	2409:40e2:24:...	2404:6800:4009:809::2001	0.089637521	TCP	114	[TCP Dup ACK 4#6] Seq=...
16	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.307218042	TLSv1.2	1294	[TCP Previous segment...]
17	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.307218408	TLSv1.2	1294	Application Data
18	2409:40e2:24:...	2404:6800:4009:809::2001	0.307352118	TCP	114	[TCP Dup ACK 4#7] Seq=...
19	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	0.354502905	TLSv1.2	1294	Application Data [T...

Ethernet is **switched communication**.

- Your PC connects to a **switch**
- The switch sends packets **only to the intended port**

WiFi is a **shared medium (broadcast)**

- All devices transmit over **air**
- Your Wireshark (especially in monitor mode) can:
 - Capture packets from **other devices**
 - See **retransmissions**

2: Traffic Clustering & Analysis

Questions

1. Separate packets into:
 - Local network (same subnet)
 - External network (internet)
2. Estimate:
 - % of packets going to **router/gateway**
 - % of packets going to **other local machines**
3. Identify:
 - Packets where **source or destination is your MAC address**
 - Packets coming from **internet to your machine**

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
						
Apply a display filter ... <Ctrl-/>						
No.	Source	Destination	Time	Protocol	Length	Info
1	172.64.148.235	10.10.106.78	0.000000000	TLSv1.2	90	Application Data
2	10.10.106.78	172.64.148.235	0.000220456	TLSv1.2	94	Application Data
3	172.64.148.235	10.10.106.78	0.028710682	TCP	66	443 → 56056 [ACK] Seq=25 Ack=29 Win=19 Le
4	10.10.106.78	142.251.221.110	0.161952049	TLSv1.2	186	Application Data
5	10.10.106.78	142.251.221.110	0.161987676	TLSv1.2	1708	Application Data, Application Data
6	142.251.221.110	10.10.106.78	0.184845600	TCP	66	443 → 34828 [ACK] Seq=1 Ack=1521 Win=976
7	142.251.221.110	10.10.106.78	0.185576660	TLSv1.2	105	Application Data
8	10.10.106.78	142.251.221.110	0.225605340	TCP	66	34828 → 443 [ACK] Seq=1763 Ack=40 Win=501
9	142.251.221.110	10.10.106.78	0.332957017	TLSv1.2	136	Application Data
10	10.10.106.78	142.251.221.110	0.332997948	TCP	66	34828 → 443 [ACK] Seq=1763 Ack=110 Win=50

Your IP = 10.10.106.78

Same subnet = local network (10.10.106.x)

10.10.106.0 → network address

10.10.106.1 → router <YOU → ROUTER → INTERNET SERVER>

10.10.106.78 → your system

10.10.106.255 → broadcast

ip.addr == 10.10.106.0/24 [PACKETS WITHIN THE LAN]

Wired

!(ip.addr == 10.10.106.0/24)						
No.	Source	Destination	Time	Protocol	Length	Info
1	Cisco_c5:4f:02	Broadcast	0.0000000000	0x8899	60	Realtek Laye
2	Cisco_57:ff:18	Broadcast	0.045737568	0x8899	60	Realtek Laye
12	Cisco_f6:3a:33	PVST+	1.096826323	STP	64	Conf. Root =
13	Cisco_61:49:39	Spanning-tree-(for-bridges)_00	1.718189614	STP	60	RST. Root =
14	Cisco_c5:4f:02	Broadcast	2.000229221	0x8899	60	Realtek Laye
15	Cisco_57:ff:18	Broadcast	2.045853106	0x8899	60	Realtek Laye
20	Cisco_f6:3a:33	PVST+	3.110045132	STP	64	Conf. Root =
64	Cisco_61:49:39	Spanning-tree-(for-bridges)_00	3.718177304	STP	60	RST. Root =
66	Cisco_c5:4f:02	Broadcast	3.999977713	0x8899	60	Realtek Laye
67	Cisco_57:ff:18	Broadcast	4.045877989	0x8899	60	Realtek Laye
70	Cisco_f6:3a:33	PVST+	5.123409692	STP	64	Conf. Root =
71	Cisco_61:49:39	Spanning-tree-(for-bridges)_00	5.718196439	STP	60	RST. Root =
72	Cisco_c5:4f:02	Broadcast	5.999994149	0x8899	60	Realtek Laye
73	Cisco_57:ff:18	Broadcast	6.045860169	0x8899	60	Realtek Laye
98	Cisco_f6:3a:33	PVST+	7.136829341	STP	64	Conf. Root =
121	Cisco_61:49:39	Spanning-tree-(for-bridges)_00	7.718343526	STP	60	RST. Root =
141	Cisco_c5:4f:02	Broadcast	8.000018432	0x8899	60	Realtek Laye

!(ip.addr == 10.10.106.0/24) [External network packets]

Wireless

!(ip.addr == 10.10.106.0/24)						
No.	Source	Destination	Time	Protocol	Length	Info
1447	2409:40e2:24:...	64:ff9b::d8ef:2074	20.528730...	TLSv1.3	125	Application Data
1448	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.531986...	TCP	86	443 → 58284 [ACK]
1449	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.531986...	TCP	86	443 → 58284 [ACK]
1450	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.563516...	TCP	1294	443 → 58284 [ACK]
1451	2409:40e2:24:...	2404:6800:4009:826::2001	20.563544...	TCP	86	58284 → 443 [ACK]
1452	10.47.70.160	173.194.154.166	20.566273...	TCP	74	[TCP Retransmission]
1453	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.586534...	TLSv1.3	1294	Server Hello, Cha...
1454	2409:40e2:24:...	2404:6800:4009:826::2001	20.586560...	TCP	86	58284 → 443 [ACK]
1455	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.613807...	TCP	1294	443 → 58284 [ACK]
1456	2409:40e2:24:...	2404:6800:4009:826::2001	20.613833...	TCP	86	58284 → 443 [ACK]
1457	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.645988...	TCP	1294	443 → 58284 [PSH, ...]
1458	2409:40e2:24:...	2404:6800:4009:826::2001	20.646031...	TCP	86	58284 → 443 [ACK]
1459	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.646048...	TCP	125	[TCP Spurious Retr...
1460	2409:40e2:24:...	2404:6800:4009:813::200e	20.646065...	TCP	98	[TCP Dup ACK 1335#...
1461	2600:1f18:24e...	2409:40e2:24:2762:5ef4:39a0:...	20.649296...	TCP	98	[TCP Dup ACK 1246#...
1462	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.654956...	TCP	86	443 → 33322 [ACK]
1463	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.654976...	TCP	94	443 → 45168 [SYN, ...]
1464	2409:40e2:24:...	2404:6800:4009:14::6	20.655018...	TCP	86	45168 → 443 [ACK]
1465	2409:40e2:24:...	2404:6800:4009:14::6	20.655591...	TLSv1	1829	Client Hello
1466	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.657369...	TCP	86	443 → 33322 [ACK]
1467	2404:6800:400...	2409:40e2:24:2762:5ef4:39a0:...	20.657369...	TLSv1.2	125	Application Data
1468	2600:1f18:24e...	2409:40e2:24:2762:5ef4:39a0:...	20.659914...	TCP	86	443 → 37886 [ACK]
1469	2600:1f18:24e...	2409:40e2:24:2762:5ef4:39a0:...	20.659914...	TCP	86	443 → 37886 [ACK]

!(ip.addr == 10.10.106.0/24) [External network packets]

```
▶ Frame 3: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface eno1, id 0
▶ Ethernet II, Src: 6c:0b:5e:56:55:06 (6c:0b:5e:56:55:06), Dst: HewlettP_ba:4a:01 (d8:94:03:ba:4a:01)
  ▶ Destination: HewlettP_ba:4a:01 (d8:94:03:ba:4a:01)
  ▶ Source: 6c:0b:5e:56:55:06 (6c:0b:5e:56:55:06)
  Type: IPv4 (0x0800)
▶ Internet Protocol Version 4, Src: 10.10.106.78, Dst: 172.64.155.209
▶ Transmission Control Protocol, Src Port: 56460, Dst Port: 443, Seq: 1, Ack: 313, Len: 0
```

Src IP: 10.10.106.78

Dst IP: 172.64.155.209

Ethernet II

Src: 6c:0b:5e:56:55:06 / YOUR system MAC

Dst: d8:94:03:ba:4a:01 / ROUTER MAC

% of packets going to router/gateway

Filter: eth.addr == 6c:0b:5e:56:55:06

Includes: Your system $\longleftrightarrow$ Internet, All traffic through the router

Packets: 332096 · Displayed: 325929 (98.1%)

$$\% = (\text{Displayed} / \text{Total}) \times 100$$

$$= (325929 / 332096) \times 100$$

$$\approx 98.1\%$$

% of packets going to other local machines

A single packet can be:

- From your system
- Going to router
- Coming from internet

Same packet counted in multiple categories

Packet: 10.10.106.78 → 172.64.155.209

This packet:

- Goes to router (MAC level)
- Goes to internet

```
ip.addr == 10.10.106.0/24 && !(ip.addr == 10.10.106.78) && !(eth.addr ==  
6c:0b:5e:56:55:06)
```

Includes packets where - Source OR destination IP is in

Excludes: Any packet involving your system

Excludes: Any packet involving router MAC

3: Protocol-Based Analysis (TCP/UDP)

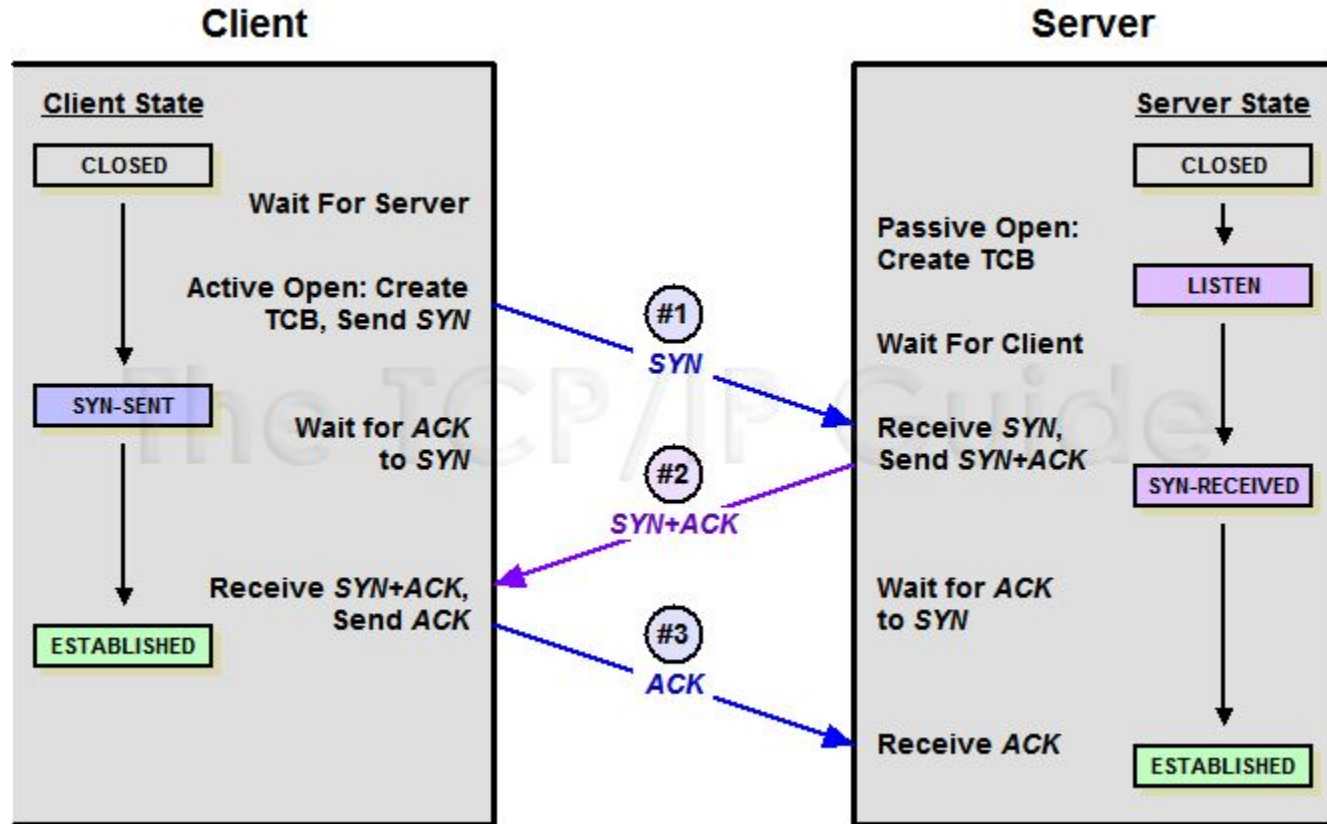
Task: Apply filters and analyze protocol behavior.

Questions

1. Using filters: tcp, udp
2. Identify:
 - One TCP connection (3-way handshake)
 - One UDP communication (e.g., DNS)
3. Explain:
 - Difference between TCP and UDP based on your observation

TCP 3 way handshake

Port:34298



Port:80

HTTP
Server

SYN (Client → Server)

tcp.flags.syn == 1 && tcp.flags.ack == 0						
No.	Source	Destination	Time	Protocol	Length	Info
305	10.10.106.78	34.0.132.167	1.108637494	TCP	74	36392 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
847	10.10.106.78	142.250.129.94	10.107288...	TCP	74	56246 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
854	10.10.106.78	142.250.129.94	10.358160...	TCP	74	56250 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
916	10.10.106.78	34.89.141.94	11.166604...	TCP	74	45970 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
1416	10.10.106.78	216.239.38.178	17.055917...	TCP	74	48808 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
1440	10.10.106.78	142.250.193.163	17.196809...	TCP	74	38870 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
2878	10.10.106.78	3.233.158.24	35.153689...	TCP	74	38226 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
2944	10.10.106.78	3.233.158.24	35.404850...	TCP	74	38234 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3114	10.10.106.78	3.233.158.24	35.652489...	TCP	74	38242 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3524	10.10.106.78	142.251.223.170	41.766758...	TCP	74	33968 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3549	10.10.106.78	142.251.223.170	41.949015...	TCP	74	33974 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3858	10.10.106.78	104.18.27.120	43.673851...	TCP	74	34298 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3859	10.10.106.78	104.18.27.120	43.673896...	TCP	74	34312 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3860	10.10.106.78	104.18.27.120	43.673978...	TCP	74	55122 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3898	10.10.106.78	178.156.202.227	43.821812...	TCP	74	50400 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3907	10.10.106.78	178.156.202.227	44.072511...	TCP	74	50414 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS
3913	10.10.106.78	178.156.202.227	44.879086...	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 50400 → 443 [SYN]
3916	10.10.106.78	178.156.202.227	45.135137...	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 50414 → 443 [SYN]
3928	10.10.106.78	178.156.202.227	45.903146...	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 50400 → 443 [SYN]

SYN = 1, ACK = 0

Filter : tcp.flags.syn == 1 && tcp.flags.ack == 0

This is the FIRST step of TCP handshake (Port:34298->80 (HTTP Server))

Client (your system) is initiating connection to server

SYN-ACK (Server → Client)

tcp.flags.syn == 1 && tcp.flags.ack == 1						
No.	Source	Destination	Time	Protocol	Length	Info
277	142.251.223.3	10.10.106.78	0.744123952	TCP	74	443 → 41896 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SA=
306	34.0.132.167	10.10.106.78	1.111084608	TCP	74	443 → 36384 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SA=
321	34.0.132.167	10.10.106.78	1.379174849	TCP	74	443 → 36392 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SA=
855	142.250.129.94	10.10.106.78	10.453764...	TCP	74	443 → 56246 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SA=
876	142.250.129.94	10.10.106.78	10.705889...	TCP	74	443 → 56250 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SA=
880	142.250.129.94	10.10.106.78	10.759226...	TCP	74	[TCP Out-Of-Order] 443 → 56246 [SYN, ACK] Seq=0 Ack=1 Win=65535
893	142.250.129.94	10.10.106.78	11.006698...	TCP	74	[TCP Out-Of-Order] 443 → 56250 [SYN, ACK] Seq=0 Ack=1 Win=65535
922	34.89.141.94	10.10.106.78	11.339553...	TCP	74	443 → 45970 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SA=
1417	216.239.38.178	10.10.106.78	17.082162...	TCP	74	443 → 48808 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SA=
1445	142.250.193.163	10.10.106.78	17.221472...	TCP	74	443 → 38870 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SA=
2945	3.233.158.24	10.10.106.78	35.425194...	TCP	74	443 → 38226 [SYN, ACK] Seq=0 Ack=1 Win=32352 Len=0 MSS=1360 SA=
3115	3.233.158.24	10.10.106.78	35.674607...	TCP	74	443 → 38234 [SYN, ACK] Seq=0 Ack=1 Win=31924 Len=0 MSS=1400 SA=
3150	3.233.158.24	10.10.106.78	35.932679...	TCP	74	443 → 38242 [SYN, ACK] Seq=0 Ack=1 Win=31668 Len=0 MSS=1230 SA=
3525	142.251.223.170	10.10.106.78	41.791930...	TCP	74	443 → 33968 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SA=
3551	142.251.223.170	10.10.106.78	41.974708...	TCP	74	443 → 33974 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SA=
3861	104.18.27.120	10.10.106.78	43.674673...	TCP	62	80 → 34298 [SYN, ACK] Seq=0 Ack=1 Win=63980 Len=0 MSS=1460 WS=1
3863	104.18.27.120	10.10.106.78	43.674673...	TCP	62	80 → 34312 [SYN, ACK] Seq=0 Ack=1 Win=63980 Len=0 MSS=1460 WS=1
3874	104.18.27.120	10.10.106.78	43.743040...	TCP	74	443 → 55122 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SA=
3993	142.251.43.131	10.10.106.78	47.704415...	TCP	74	443 → 59740 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SA=

SYN = 1, ACK = 1

Filter : tcp.flags.syn == 1 && tcp.flags.ack == 1

This is the SECOND step of TCP handshake (Port:80->34298)

Client (your system) is initiating connection to server

ACK (Client → Server)

tcp.flags.ack==1 && tcp.len==0						
No.	Source	Destination	Time	Protocol	Length	Info
3831	142.251.43.131	10.10.106.78	43.005396...	TCP	66	[TCP ACKed unseen segment] 443 → 39076 [FIN, ACK] Seq=74 Ack=...
3832	10.10.106.78	142.251.43.131	43.005431...	TCP	66	39076 → 443 [ACK] Seq=3 Ack=75 Win=478 Len=0 TSval=115729422...
3835	142.251.222.174	10.10.106.78	43.113198...	TCP	66	443 → 45574 [ACK] Seq=1623 Ack=4004 Win=860 Len=0 TSval=29910...
3836	142.251.222.174	10.10.106.78	43.113558...	TCP	66	443 → 45574 [ACK] Seq=1623 Ack=5936 Win=853 Len=0 TSval=29910...
3838	10.10.106.78	142.251.222.174	43.155062...	TCP	66	45574 → 443 [ACK] Seq=5936 Ack=1662 Win=468 Len=0 TSval=13635...
3840	10.10.106.78	142.251.222.174	43.212253...	TCP	66	45574 → 443 [ACK] Seq=5936 Ack=1732 Win=468 Len=0 TSval=13635...
3842	10.10.106.78	142.251.222.174	43.212764...	TCP	66	45574 → 443 [ACK] Seq=5936 Ack=2362 Win=464 Len=0 TSval=13635...
3844	10.10.106.78	142.251.222.174	43.213198...	TCP	66	45574 → 443 [ACK] Seq=5936 Ack=2655 Win=462 Len=0 TSval=13635...
3846	10.10.106.78	142.251.222.174	43.213207...	TCP	66	45574 → 443 [ACK] Seq=5936 Ack=2686 Win=462 Len=0 TSval=13635...
3848	10.10.106.78	142.251.222.174	43.213213...	TCP	66	45574 → 443 [ACK] Seq=5936 Ack=2725 Win=462 Len=0 TSval=13635...
3851	142.251.222.174	10.10.106.78	43.239458...	TCP	66	443 → 45574 [ACK] Seq=2725 Ack=6010 Win=853 Len=0 TSval=29910...
3861	104.18.27.120	10.10.106.78	43.674673...	TCP	62	80 → 34298 [SYN, ACK] Seq=0 Ack=1 Win=63980 Len=0 MSS=1460 WS...
3862	10.10.106.78	104.18.27.120	43.674731...	TCP	54	34298 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0
3863	104.18.27.120	10.10.106.78	43.674673...	TCP	62	80 → 34312 [SYN, ACK] Seq=0 Ack=1 Win=63980 Len=0 MSS=1460 WS...
3864	10.10.106.78	104.18.27.120	43.674739...	TCP	54	34312 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0
3865	216.239.34.223	10.10.106.78	43.682831...	TCP	66	[TCP ACKed unseen segment] 443 → 58420 [ACK] Seq=1 Ack=199 W...
3870	216.239.34.223	10.10.106.78	43.699721...	TCP	66	443 → 58420 [ACK] Seq=40 Ack=355 Win=1118 Len=0 TSval=154354...
3871	10.10.106.78	216.239.34.223	43.723072...	TCP	66	58420 → 443 [ACK] Seq=355 Ack=40 Win=1794 Len=0 TSval=598815...
3872	34.120.52.64	10.10.106.78	43.727304...	TCP	66	443 → 57970 [ACK] Seq=509 Ack=556 Win=1039 Len=0 TSval=48633...
3873	34.120.52.64	10.10.106.78	43.737973...	TCP	66	443 → 57970 [ACK] Seq=509 Ack=705 Win=1039 Len=0 TSval=48633...
3874	104.18.27.120	10.10.106.78	43.743040...	TCP	74	443 → 55122 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 S...
3875	10.10.106.78	104.18.27.120	43.743089...	TCP	66	55122 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=91987864...
3878	10.10.106.78	216.239.34.223	43.813348...	TCP	66	58420 → 443 [ACK] Seq=355 Ack=113 Win=1794 Len=0 TSval=598815...
3880	10.10.106.78	216.239.34.223	43.813668...	TCP	66	58420 → 443 [ACK] Seq=355 Ack=702 Win=1816 Len=0 TSval=598815...

ACK = 1 ::Filter : tcp.flags.ack == 1 && tcp.len == 0

This is the THIRD step of TCP handshake (Port:34298->80)

Client (your system) is initiating connection to server

Close and check for FIN

UDP

dns						
No.	Source	Destination	Time	Protocol	Length	Info
110	36.50.50.50	10.10.106.78	1.048504725	DNS	116	Standard query response 0x20e1 A addons-pa.clients6.goc
111	36.50.50.50	10.10.106.78	1.052587415	DNS	150	Standard query response 0x7ebf HTTPS addons-pa.clients6
302	10.10.106.78	36.50.50.50	3.910185280	DNS	86	Standard query 0x8e05 A www.youtube.com OPT
303	10.10.106.78	36.50.50.50	3.910230613	DNS	86	Standard query 0x7cde HTTPS www.youtube.com OPT
304	10.10.106.78	36.50.50.50	3.910267671	DNS	91	Standard query 0x6189 A fonts.googleapis.com OPT
305	10.10.106.78	36.50.50.50	3.910295738	DNS	91	Standard query 0xa6ce HTTPS fonts.googleapis.com OPT
306	10.10.106.78	36.50.50.50	3.910321340	DNS	88	Standard query 0x208e A fonts.gstatic.com OPT
310	10.10.106.78	36.50.50.50	3.926413724	DNS	92	Standard query 0xa1e2 A events.givefreely.com OPT
311	10.10.106.78	36.50.50.50	3.926461438	DNS	92	Standard query 0x1e5c HTTPS events.givefreely.com OPT
312	36.50.50.50	10.10.106.78	3.939737919	DNS	135	Standard query response 0x7cde HTTPS www.youtube.com CN
313	36.50.50.50	10.10.106.78	3.940106275	DNS	104	Standard query response 0x208e A fonts.gstatic.com A 17
316	10.10.106.78	36.50.50.50	3.940856876	DNS	82	Standard query 0xd9fb A i.ytimg.com OPT
317	10.10.106.78	36.50.50.50	3.940887503	DNS	82	Standard query 0x826d HTTPS i.ytimg.com OPT
318	36.50.50.50	10.10.106.78	3.942050949	DNS	107	Standard query response 0x6189 A fonts.googleapis.com A
319	36.50.50.50	10.10.106.78	3.944190092	DNS	148	Standard query response 0xa6ce HTTPS fonts.googleapis.c
320	36.50.50.50	10.10.106.78	3.944714532	DNS	376	Standard query response 0x8e05 A www.youtube.com CNAME
326	36.50.50.50	10.10.106.78	3.970043773	DNS	139	Standard query response 0x826d HTTPS i.ytimg.com SOA ns
327	36.50.50.50	10.10.106.78	3.972941456	DNS	242	Standard query response 0xd9fb A i.ytimg.com A 172.217.
328	36.50.50.50	10.10.106.78	3.973000000	DNS	100	Standard query response 0x1e5c HTTPS events.givefreely.com OPT

ASSIGNMENT 1: HTTP Sniffing & Security Awareness

Instructions

1. Capture traffic while opening: <http://testaspnet.vulnweb.com/login.aspx>
2. Apply filter: http

Questions

1. Identify:
 - HTTP request packet
 - HTTP response packet
2. Extract:
 - Requested URL
 - Server response code

In POST we have a packet with form data click on the packet with user info and the application URL encoded

```
▼ HTML Form URL Encoded: application/x-www-form-urlencoded
  ▶ Form item: "__EVENTTARGET" = ""
  ▶ Form item: "__EVENTARGUMENT" = ""
  ▶ Form item: "__VIEWSTATE" = "/wEPDwUKLTIyMzk2OTgxMQ9kFgICAQ9kFgICAQ9kFgQCAQ8WBB4EaHJlZgUKbG9
  ▶ Form item: "__VIEWSTATEGENERATOR" = "C2EE9ABB"
  ▶ Form item: "__EVENTVALIDATION" = "/wEWwWLoz/fGCgLStq24BwK3jsrkbALtuvfLDQKC3IeGDAKuztjRBQKA1
  ▶ Form item: "tbUsername" = "testlog"
  ▶ Form item: "tbPassword" = "123"
  ▶ Form item: "cbPersistCookie" = "on"
  ▶ Form item: "btnLogin" = "Login"
```

ASSIGNMENT 2: Sniffing based script

Capture network traffic during a login process (HTTP and HTTPS) and save it as a single PCAP/PCAPNG file. Perform all further analysis using scripts on this captured file. Identify where login credentials are present and evaluate their visibility. Observe that HTTP allows direct extraction of credentials, whereas HTTPS encrypts the payload. Finally, demonstrate brute-force password recovery in a controlled hash-based scenario.

Capture Once (PCAP File Generation)

- Use Wireshark to capture login activity (HTTP + HTTPS)
- Save the capture as a `.pcap` / `.pcapng` file

HTTP Credential Extraction (Using Script)

- Parse the PCAP file using a script
- Identify HTTP POST packets
- Extract username and password from payload

HTTPS Analysis (Using Script)

- Parse the same PCAP file
- Identify TLS packets
- Attempt to locate credentials and observe encrypted payload

Brute-force Demonstration (TOOL/Script-Based)

- Use a Password Recovery Tool (OR)
- Assume a 2-digit numeric password (00–99) / generate a simple password file.
- Generate all combinations
- Hash and compare with stored hash to recover password

